

SUMMARY

AI Engineer with 7+ years of experience designing & delivering end-to-end AI and data platforms. Skilled across the full spectrum of AI, from traditional machine learning (predictive modeling, AutoML, & MLOps) to the latest generative & agentic AI (LLM-powered agents, RAG, & Graph-RAG). Strong foundation in large-scale data engineering, adept at taking solutions from data modeling & experimentation to scalable, secure production. Passionate about democratizing AI & drive innovation.

EDUCATION

Georgia Institute of Technology – M.S. Computer Science (Spec. ML/AI), GPA: 3.9/4.0	2023 – 2026
NIT Bhopal – B.Tech Computer Science & Engineering, CGPA: 8.17/10	2015 – 2019
Kendriya Vidyalaya (CBSE) – Class 12 th : 95.4%, Class 10 th : 98.6%	2004 – 2015

EXPERIENCE

Microsoft

SDE2 (L62) - Data Engineering/ Data Science / Agentic AI

India

2019 – Present

- **Account Researcher AI Agent Platform:** Architected a production LLM-powered agent converting natural-language sales queries into grounded insights, using a scalable FastAPI microservice. Engineered an agentic orchestration loop with query decomposition, parallel LLM calls and NL2SQL grounded via RAG over Azure AI Search, driving accuracy with iterative prompt tuning, semantic validation, & self-correction against business rules. Optimized latency & token cost via semantic caching, async concurrency, & SQL connection pooling; hardened with row-level security, & telemetry tracing. Deployed on Azure & piloted to 3k+ sellers across Microsoft globally.
- **Customer Graph-RAG:** Built an end-to-end Graph-RAG platform transforming Microsoft's global enterprise customer data across multiple domains (Account, Opportunity, Customer Health, and Azure Consumed Revenue etc) into a unified knowledge graph that answers natural-language questions with grounded LLM responses and a live subgraph visualization for explainability. Architected a three-stage pipeline: offline graph build using Fabric Graph, in-memory load, and real-time serving via a semantic retrieval engine over Azure OpenAI embeddings.
- **AutoBrewML:** Created an open-source Microsoft framework that automates the end-to-end machine learning workflow (data prep, model training, & deployment) on Azure, letting teams build models with 80% less manual effort & no prior ML expertise. Published on PyPI (~2,000 downloads in the first month), presented as lead speaker at Microsoft's internal Machine Learning, AI & Data Science (MLADS) conference, and filed a patent for its data-sampling module, later reused by the Microsoft Cloud Data Science team to speed up incident debugging on smaller data samples.
- **MLOps / AIOps Platform:** Led the design & build of a unified MLOps & AIOps platform on Microsoft Fabric that gives teams a standardized, secure way to ship both classical ML models and LLM-powered agents end-to-end, automating model training, versioning, monitoring, deployment, data-drift detection, and retraining for classical ML while operationalizing generative & agentic AI through prompt/version management, LLM evaluation and guardrails, RAG pipeline orchestration, and continuous quality, latency, cost, and hallucination telemetry.
- **Predictive Resource Forecasting:** Built a machine learning model that forecasts each consultant's weekly billable hours at 75–80% accuracy, replacing a manual planning process for Project Managers (whose compensation is tied to planning accuracy); adopted globally after a successful pilot.
- **Automated Data Tagging:** Built a recommendation engine that automatically assigns business-friendly labels to newly ingested data columns across a large commercial data platform (Sales, Consumption, Marketing, Finance, & Professional Services), using a hybrid approach combining keyword-frequency (TF-IDF) & word-embedding techniques to save significant manual engineering effort.
- **Enterprise Data Warehousing & BI:** Designed & maintained multiple large-scale data warehouses & ETL processes that consolidated data from multiple sources into a single 1M+ user platform, delivering business intelligence and reporting for a multi-million-dollar Microsoft Consulting Services solution.

LEADERSHIP & COMMUNITY OUTREACH

- **Diversity & Inclusion Lead, Microsoft India:** Drove diversity & inclusion for the Digital Professional Services org in India, launching a sustainable, non-conventional program alongside senior leadership, HR, & Legal. Improved D&I scores in the company-wide employee sentiment (Pulse) survey, with the initiative featured in the India Development Center newsletter for its unique approach.
- Coordinated external outreach programs for underprivileged girls and women with the **Microsoft India Philanthropy** team & the **Women@Microsoft External Outreach** pillar

TECHNICAL SKILLS

Focus Areas: Machine Learning & Modelling (MLOps), Generative / Agentic / NLP Solutions (AIOps), RAG & Graph-RAG, Data Modelling, Database Development, Business Intelligence (BI), Azure Cloud Infrastructure

Languages & Frameworks: Python, SQL, Scala, PyTorch, TensorFlow, Scikit-learn, Pandas, Spark, PySpark

Platforms & Tools: Azure OpenAI, Azure AI Search, Azure Cognitive Services, Microsoft Fabric, Azure Data Factory, Azure Data Lake, Azure Synapse, Azure Databricks, Power BI